Category Theory for Mathematical Physics — Exercise Sheet 14

Topological quantum field theory: a synthesis

Variant: Questions only

Conventions as in Lecture 14: time runs upward; a Frobenius algebra is $(A, \mu, \iota, \Delta, \varepsilon_A)$, drawn with the pieces of § 14.2 (the pants μ and Δ, the discs ι and ε_A); P is the flip of Vect, $w = \mu \circ \Delta \circ \iota$ the handle element, Σ_g the closed surface of genus $g \geq 0$, and $Z(\Sigma_g) = \varepsilon_A(w^g)$. Groups are multiplicative, with identity 1, and $[a, b] = aba^{-1}b^{-1}$. In this sheet $\beta = \varepsilon_A \circ \mu$ and $\gamma = \Delta \circ \iota$. In Exercise 4, n_0, n_1, n_2 count vertices, edges and faces, and Z_1, B_1 are spaces of chains (not partition functions or centres); in the starred exercise $N(x)$ is a count.

Exercise 1 (The Frobenius axioms as surfaces). (i) Draw as surfaces, glued from the pieces of Lecture 14, the associativity of μ, the unit law $\mu \circ (\iota \otimes \mathrm{id}) = \mathrm{id}$ and the Frobenius relation Equation (14.3). Which pictures give coassociativity and the counit law? Which diffeomorphism of the pair of pants gives commutativity?

(ii) Show that $(\mathrm{id} \otimes \beta) \circ (\gamma \otimes \mathrm{id}) = \mathrm{id}_A = (\beta \otimes \mathrm{id}) \circ (\mathrm{id} \otimes \gamma)$, draw it as in Equation (14.1), and deduce that β is a nondegenerate pairing with $\beta(xy, z) = \beta(x, yz)$.

Exercise 2 (The handle element). (i) Let $A = \mathbb{C}[x]/(x^2)$ with $\varepsilon_A(1) = 0$, $\varepsilon_A(x) = 1$. Show that β is nondegenerate, find $\Delta(1)$ and $\Delta(x)$ from $\Delta(a) = \sum_i ae_i \otimes e^i$, where $\{e_i\}$ and $\{e^i\}$ are dual bases, $\beta(e_i, e^j) = \delta_{ij}$, and compute w and $Z(\Sigma_g)$ for every g.

(ii) For any commutative Frobenius algebra show that $Z(T^2) = \varepsilon_A(w) = \dim A$: the case $\Sigma = S^1$ of Theorem 14.2(ii).

(iii) Let $A = \mathbb{C}[\mathbb{Z}_2]$ with $\varepsilon_A(g) = \delta_{g,1}$. Compute $Z(\Sigma_g)$. What does the counit $\varepsilon_A(f) = f(1)/|G|$ of Theorem 14.6 give instead?

Exercise 3 (Dijkgraaf–Witten for $\mathbb{Z}_2$ and S_3). Use

$$\pi_1(\Sigma_g) = \langle a_1, b_1, \ldots, a_g, b_g \mid [a_1, b_1] \cdots [a_g, b_g] \rangle.$$

(i) For $G = \mathbb{Z}_2$, count $\#\,\mathrm{Hom}(\pi_1\Sigma_g, G)$, and check both expressions of Theorem 14.6.

(ii) For $G = S_3$, compute $Z(\Sigma_g)$ for $g = 0, 1, 2$ from the character sum, and $\#\,\mathrm{Hom}(\pi_1\Sigma_2, S_3)$. Check $g = 0$ and $g = 1$ against the count of homomorphisms directly.

Exercise 4 (The Levin–Wen count for $\mathrm{Vec}_{\mathbb{Z}_2}$). Let Γ be a connected trivalent graph in Σ_g whose faces are discs, with n_0 vertices, n_1 edges and n_2 faces, so that $n_0 - n_1 + n_2 = 2 - 2g$. The string net of $\mathrm{Vec}_{\mathbb{Z}_2}$ puts a qubit, a label 0 or 1, on each edge; $Q_v = 1$ when the labels at v add up to 0 mod 2, and $B_p = \frac{1}{2}(1 + X_p)$, where X_p adds ∂p, the boundary of the face p mod 2, to the labels (an edge along which ∂p runs twice is not flipped) (§ 14.4, with every $d = 1$ and every F-symbol 1).

(i) Let Z_1 be the closed $\mathbb{Z}_2$-chains (the label sets with $Q_v = 1$ at every v) and B_1 the sums of face boundaries. Show that the ground states are the functions on Z_1 that are constant on the cosets of B_1.

(ii) Show that $\dim Z_1 = n_1 - n_0 + 1$ and $\dim B_1 = n_2 - 1$ over $\mathbb{Z}_2$, and deduce that there are 2^{2g} ground states.

(iii) Compare the torus with Theorem 14.8(ii)–(iii) and with *Quantum Computing*. How many torus states does Theorem 14.8 give for Vec_{S_3}, and why is that not the $Z(T^2) = 3$ of Exercise 3?

Exercise 5 (Turaev–Viro, Reshetikhin–Turaev and the anomaly). (i) Give $\dim Z_{\mathrm{TV},C}(T^2)$ for $C = \mathrm{Vec}_{\mathbb{Z}_2}$, Vec_{S_3}, Fibonacci and Ising.

(ii) For Fibonacci, Ising, $\mathrm{SU}(2)_2$, the toric code and $\mathrm{Fib} \boxtimes \mathrm{Fib}^{\mathrm{rev}}$, find c mod 8 (Lecture 10) and say whether the gluing phase $e^{2\pi i c/8}$ of RT is 1.

(iii) Why is no Turaev–Viro theory equal to $\mathrm{RT}_{\mathrm{Fib}}$?

Exercise 6 ⋆. *Mednykh's formula.* Let G be a finite group, K its conjugacy classes, and for $x \in G$ let

$$\mathcal{N}(x) = \#\{(a,b) \in G^2 : [a,b] = x\}.$$

You may use, for the irreducible characters, $1_K = \frac{|K|}{|G|} \sum_\chi \overline{\chi(K)}\,\chi$ (the indicator function of the class), the relation $\sum_K |K|\ |\chi(K)|^2 = |G|$, and $\chi * \psi = \delta_{\chi\psi}\,(|G|/\chi(1))\,\chi$ for the convolution of Theorem 14.6.

(i) Show that $\mathcal{N} = \sum_K \frac{|G|}{|K|}\, 1_{K^{-1}} * 1_K$.

(ii) Deduce that $\mathcal{N}(x) = |G| \sum_\chi \chi(x)/\chi(1)$.

(iii) Show that $\#\,\mathrm{Hom}(\pi_1\Sigma_g, G) = \mathcal{N}^{*g}(1)$, and conclude that $\#\,\mathrm{Hom}(\pi_1\Sigma_g, G)/|G| = \sum_\chi (|G|/\chi(1))^{2g-2}$.